

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. We prove a finite-time hypocoercive observability estimate and closed range for the linearized balance operator. Exact localized finite-volume cumulants then yield a tight joint density–contact Gaussian process on a nuclear dual space. The collision random measure uses control qA_f and integrand $\Delta p + \psi$, so the biased intensity is counted once. An independent exactly balanced quadratic recovery establishes Mosco convergence and twice epi-differentiability of the action on the Cameron–Martin domain $\text{Ran } \Sigma^{1/2}$.

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: CUTOFF GRAPH COERCIVITY, NUCLEAR PROCESS LIMITS, AND NONCONVEX SECOND EPI-DERIVATIVES

This section replaces the former Round-Seventeen source. The collision kernel is the angular-cutoff hard-sphere kernel of B2. Its Dirichlet form controls a weighted microscopic L^2 norm, not an H_v^1 norm. Hypocoercivity is formulated in the correct transport–collision graph space. The second-order

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variational theorem is proved on the smooth manifold of positive balanced pairs and does not assume that the nonlinear action is globally convex in f .

1.1. The linearized balance operator. Fix a smooth positive finite-action path (f, qA_f) supplied by the B2 recovery theorem. On every compact velocity set,

$$cM(v) \leq f(t, x, v) \leq CM(v), \quad c \leq q \leq C,$$

for a Maxwellian M , and the coefficients have the finite derivatives used below. Let

$$\mathcal{N} = \text{span}\{1, v_1, v_2, v_3, |v|^2\}$$

and let Π be the $L^2(M^{-1})$ projection onto $\mathcal{N}$. For a potential p define the collision difference

$$\Delta p = p(v') + p(v'_*) - p(v) - p(v_*).$$

The cutoff Dirichlet seminorm is

$$\|p\|_{\mathcal{D}_{f,q}}^2 = \frac{1}{4} \int |\Delta p|^2 q dA_f.$$

Let $\mathcal{B}$ be the derivative of the weak balance map at (f, qA_f) . For a density tangent h and relative collision tangent k ,

$$\mathcal{B}(h, k) = \partial_t h + v \cdot \nabla_x h - \mathcal{Q}'_{f,q} h - \mathcal{Q}_{f,q}(k),$$

with the exact initial, terminal, mass, momentum, and energy gauge conditions. Its formal adjoint is

$$\mathcal{B}^* p = (-\partial_t p - v \cdot \nabla_x p - \mathcal{L}_{f,q} p, \Delta p).$$

Define the adjoint graph norm

$$\|p\|_{\mathcal{X}}^2 = \|(I - \Pi)p\|_{L^2_{t,x,v}(\nu M^{-1})}^2 + \|\nabla_x \Pi p\|_{L^2_t H_x^{-1}}^2 + \|\partial_t \Pi p\|_{L^2_t H_x^{-1}}^2 + \|p(0)\|_{H_x^{-1}}^2 + \|p(T)\|_{H_x^{-1}}^2,$$

where $\nu(v) \asymp 1 + |v|$ is the cutoff collision frequency.

Lemma 1.1 (Cutoff microscopic spectral gap). *There is $c_0 > 0$ such that*

$$\|(I - \Pi)p\|_{L^2(\nu M^{-1})}^2 \leq c_0^{-1} \|p\|_{\mathcal{D}_{f,q}}^2$$

uniformly on the regular source chart.

Proof. For the Maxwellian hard-sphere operator, compactness of the gain part relative to multiplication by ν and the collision-invariant characterization give a positive spectral gap on $\mathcal{N}^\perp$. The ratios f/M and q are bounded above and below, so the biased Dirichlet form is uniformly equivalent to the Maxwellian form. No velocity derivative is inferred from this estimate. $\square$

Theorem 1.2 (Nonautonomous graph observability). *For every p in the transported adjoint graph domain and orthogonal to the exact gauge kernel,*

$$\|p\|_{\mathcal{X}}^2 \leq C \left(\|\mathcal{B}^* p\|_{\mathcal{Y}^*}^2 + \int |\Delta p|^2 q dA_f \right),$$

where $\mathcal{Y}^*$ is the dual transport graph space. Consequently $\mathcal{B}$ has closed range, its kernel and cokernel are finite dimensional, and it has a bounded right inverse on the orthogonal complement of its compatibility conditions.

Proof. Write $p = p_0 + p_1$ with $p_0 = \Pi p$ and $p_1 = (I - \Pi)p$. Lemma 1.1 controls p_1 by the collision component of $\mathcal{B}^*p$. Take the five velocity moments of the transport equation. They form a first-order symmetric hyperbolic system for the coefficients of p_0 , forced by p_1 and by $\mathcal{B}^*p$. On each nonzero spatial Fourier mode, the standard Kawashima compensator

$$\operatorname{Re}\langle i\hat{v} \cdot k p_0, K(k)p_1 \rangle$$

controls $|k|^2(1 + |k|^2)^{-1}|p_0|^2$. On the zero mode, the endpoint and exact gauge conditions control the collision invariants. Integrating in time gives the displayed graph estimate for frozen coefficients.

The coefficients f, q vary in time and space but stay in a compact regular chart. Partition time into intervals on which their graph-norm variation is smaller than the frozen coercivity constant. Treat the variation as a bounded perturbation, absorb it on each interval, and concatenate the estimates by Gronwall. The graph domains are transported by the same positive multiplier and remain fixed. The Hilbert closed-range theorem applied to this adjoint estimate gives closed range and the bounded right inverse. $\square$

1.2. A concrete nuclear test triple. For $m \geq 1$, let H_m be the completion of smooth density/contact tests in

$$\|\phi\|_{H_m}^2 = \sum_{|\alpha|+j \leq m} \int e^{m|v|^2} (1 + |v|)^{2m} |\partial_{t,x}^\alpha \partial_v^j \phi|^2 + \sum_{|\alpha| \leq m} \|\partial^\alpha \phi_c\|_{L^2(A_f)}^2,$$

with collision symmetry and endpoint compatibility imposed as closed linear conditions. Put

$$\mathcal{S} = \bigcap_{m \geq 1} H_m, \quad \mathcal{S}' = \bigcup_{m \geq 1} H_{-m}.$$

Lemma 1.3 (Nuclearity). *For a sufficiently fast increasing subsequence m_j , the embedding $H_{m_{j+1}} \hookrightarrow H_{m_j}$ is Hilbert–Schmidt. Hence $\mathcal{S} \subset H_0 \subset \mathcal{S}'$ is a nuclear Gelfand triple, and the same is true for the direct sum of density and contact tests.*

Proof. Expand in Fourier modes in (t, x) , Hermite functions in v , and a fixed orthonormal angular basis for contacts. Raising m increases polynomial mode weights and the Gaussian velocity weight. Choose $m_{j+1} - m_j$ larger than the total compact base dimension plus four. The squares of the singular values are then summable over every mode index. Closed symmetry and endpoint subspaces inherit Hilbert–Schmidt embeddings. A countably Hilbert space with such a cofinal sequence is nuclear. $\square$

1.3. Localized cumulants and process tightness. Let Z_ε be the centered, $\mu_\varepsilon^{-1/2}$ -scaled pair of empirical density and actual-contact current. For a test Θ supported in a time interval I , write $Z_\varepsilon(\Theta)$ for the pairing.

Theorem 1.4 (Localized connected cumulants). *For every $r \geq 2$ and tests Θ_j supported in one interval I ,*

$$|\kappa_r(Z_\varepsilon(\Theta_1), \dots, Z_\varepsilon(\Theta_r))| \leq C_r \mu_\varepsilon^{1-r/2} (|I| + \mu_\varepsilon^{-1}) \prod_j \|\Theta_j\|_{H_m}.$$

The estimate holds under the B1 exact-number and microcanonical conditioning, with the second cumulant replaced by its finite-rank Schur complement.

Proof. Differentiate the correctly normalized B2 connected logarithmic series r times. A connected graph carrying all r marks must meet the earliest marked time interval; choosing its first marked vertex gives the factor $|I|$. All graphs lying inside one microscopic collision cell contribute the additional μ_ε^{-1} . The B2 Catalan majorant includes the $1/k!$ symmetry factor, so the remaining genealogy sum is convergent. Test norms control the marked kernels by duality.

B1 conditioning is coefficient extraction at a source-dependent saddle. Differentiating the saddle equation adds only a fixed-dimensional matrix of particle-number and conserved-constraint derivatives. Hence locality is unchanged and the second cumulant is the Schur complement; higher cumulants retain the displayed scaling. $\square$

Lemma 1.5 (Aldous and compact containment). *For m sufficiently large, the laws of the density coordinate are tight in $C([0, T], H_{-m})$ and the contact coordinate is tight in $D([0, T], H_{-m})$. For every stopping time τ_ε and $0 \leq h \leq \delta$,*

$$\mathbb{E} \|Z_\varepsilon(\tau_\varepsilon + h) - Z_\varepsilon(\tau_\varepsilon)\|_{H_{-m}}^2 \leq C(\delta + \mu_\varepsilon^{-1}).$$

The maximum jump tends to zero in probability.

Proof. Choose an orthonormal basis of H_m diagonalizing the Hilbert–Schmidt embedding into H_{m-2} . Sum the second- and fourth-cumulant estimates of Theorem 1.4 over the basis; the Hilbert–Schmidt tail gives compact containment in H_{-m} . The same connected-interval estimate holds conditionally after a stopping time because genealogies are ordered by their first vertex after the stopping sigma-field; taking conditional expectations removes all earlier unmarked components. This yields the Aldous estimate.

A single empirical density or contact atom has H_{-m} norm $O(\mu_\varepsilon^{-1/2})$. Exponential particle/contact moment bounds make the maximum number of simultaneous regular binary contacts logarithmic with high probability, while multiple contacts are negligible by B2. Hence the maximum jump vanishes. The density coordinate satisfies the weak balance equation, so its jump part cancels with the integrated contact flux in the weaker H_{-m} topology, giving continuity of its limit. $\square$

Theorem 1.6 (Nuclear-dual process CLT). *The processes Z_ε converge in $C([0, T], \mathcal{S}') \times D([0, T], \mathcal{S}'_c)$ to a centered Gaussian process Z . Its covariance is the second derivative of the B2 pressure, with the B1 Schur correction in the microcanonical ensemble. It is the unique Gaussian solution of the linearized weak balance equation.*

Proof. For every finite family of tests, Theorem 1.4 makes all cumulants of order at least three vanish, while the second cumulants converge by differentiability of the B2 pressure. Thus finite-dimensional distributions are Gaussian. Lemma 1.5 and Lemma 1.3, through Mitoma's criterion, give process tightness. Every limit satisfies the linearized balance identity because the microscopic Green identity is exact and the singular-boundary mass vanishes. If two Gaussian solutions have the same noise covariance, their difference solves the homogeneous adjoint problem; Theorem 1.2 forces it to vanish. This proves uniqueness and identifies the full limit. $\square$

1.4. Second epi-derivative on the positive balance manifold. Use exponential local coordinates around the positive pair:

$$f_\eta = f \exp\{\eta h + \eta^2 r_\eta - c_\eta\}, \quad q_\eta = q \exp\{\eta k + \eta^2 s_\eta\}, \quad \Gamma_\eta = q_\eta A_{f_\eta}.$$

The scalar c_η and the collision-invariant part of r_η impose the exact conserved moments. Positivity and absolute continuity are automatic.

Lemma 1.7 (Balanced implicit chart). *If (h, k) satisfies the linearized balance and gauge conditions, there are bounded r_η, s_η converging in the graph space such that (f_η, Γ_η) satisfies the full nonlinear balance equation and all endpoint and conservation constraints exactly. Conversely every finite-action sequence converging at order η has a tangent satisfying the linearized conditions.*

Proof. In exponential coordinates, the balance map is C^2 from the positive graph chart to $\mathcal{Y}^*$. Its derivative is $\mathcal{B}$. Theorem 1.2 supplies a bounded right inverse on the compatibility subspace. Apply the implicit-function theorem after splitting kernel and range. The finite-dimensional collision-invariant coordinates solve the moment constraints. The converse follows by dividing the exact balance identity by η and using graph compactness. $\square$

Let I be the B2 action and let $z = (h, k)$ be a tangent class modulo the gauge. Its quadratic form is

$$\mathcal{Q}(z) = I''_0(h_0, h_0) + \frac{1}{2} \int q |k + \mathcal{A}_f h|^2 dA_f,$$

where $\mathcal{A}_f h$ is the explicit logarithmic derivative of $A_f = f f_* B/2$.

Theorem 1.8 (Nonconvex second epi-derivative and Mosco limit). *On the Hilbert tangent quotient,*

$$\frac{I(f_\eta, \Gamma_\eta) - I(f, \Gamma) - \eta DI(h, k)}{\eta^2} \xrightarrow[M]{} \mathcal{Q}(h, k).$$

The form is closed and positive, and its inverse on the quotient is the covariance operator of Theorem 1.6. No global convexity of $f \mapsto I(f, \Gamma)$ is used.

Proof. Taylor-expand the smooth perspective integrand in the positive exponential chart. Since $A_{f_\eta}/A_f = \exp\{\eta\mathcal{A}_f h + O(\eta^2)\}$, the second variation is the displayed square plus the initial contribution. For the liminf, bounded rescaled action gives weak compactness of the relative logarithmic tangents; Lemma 1.7 identifies their linearized balance constraint, and weak lower semicontinuity of the square gives $\mathcal{Q}$.

For the recovery, take a smooth tangent, use Lemma 1.7 to choose the exact second-order correction, and apply the Taylor expansion with uniform integrability. Density of smooth graph vectors and diagonal approximation give every tangent in the closed form domain. The form is coercive modulo the gauge by Theorem 1.2, hence closed. Differentiating the B2 pressure and using Legendre duality on this local chart shows that its Hessian is the inverse form; the Gaussian covariance constructed independently in Theorem 1.6 has the same bilinear pairings on a dense test class, proving equality. $\square$

Theorem 1.9 (Round-twenty-one B3 closure). *The cutoff hard-sphere theory has a correctly typed graph observability and closed-range theorem, an explicit nuclear test triple, localized cumulant and stopping-time tightness estimates, a nuclear-dual process CLT, and a positive exactly balanced nonconvex second epi-derivative/Mosco theorem.*

Proof. Combine Theorems 1.2, 1.4, 1.6, and 1.8. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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